

Solving $Ax = 0$: Pivot Variables, Special Solutions

Lesson 6 said what the do-nothing trades are. This lesson is a four-move recipe that finds every one of them, for any menu of funds, even when there are more funds than rules.

PART 1 Index

One question runs through the whole lesson: **which trades change nothing?** Each topic is one move of the recipe that answers it.

1. The question, and a guess

Four funds, three rules. We can guess two do-nothing trades. Are there more? The recipe will tell us for sure.

2. Move 1: simplify the rules

Elimination, explained as rewriting the rules ("return beyond 12%"). Why that can't change the do-nothing trades.

3. Move 2: pivot columns and free columns

You pick two amounts, the rules force the other two. Why it's the ETF and balanced you pick. Rank.

4. Move 3: the special solutions

Choose the free amounts one at a time. Out come exactly our two guesses.

5. Move 4: every do-nothing trade

Every do-nothing trade is a mix of the special solutions, and nothing else exists. With a try-it box.

6. The same four moves on smaller menus

A 2×2 and two 3×3 s. How many special solutions: $n - r$.

7. The shortcut: the reduced form R

Two more row operations, and the special solutions can be read straight off.

METHODS IN THIS LESSON

Row operations keep $N(A)$

Echelon form U

Pivot and free variables

Rank r

Special solutions

$n - r$ rule

Reduced row echelon form R

Nullspace matrix N

PART 2 Prerequisites

Click any link to review it, then come back.

- The nullspace (Lesson 6): the trades with $Ax = 0$, which change nothing.
- Elimination, 3×3 and back substitution (Lesson 2).
- Gauss–Jordan (Lesson 3): clearing above the pivots as well as below.

PART 3 Explanation

THE WHOLE LESSON IN ONE LINE

To find every trade that changes nothing: (1) simplify the rules by elimination, (2) see which fund amounts you can choose freely, (3) choose them one at a time to get a few basic do-nothing trades, (4) mix those to get all of them.

1. The question, and a guess

In Lesson 6 a menu of three funds had one do-nothing trade, and we found it by hand. Real menus are bigger: many products, a few things you measure. So we need a recipe that works for **any** menu. Here is the menu we'll use for the whole lesson.

THE MENU: FOUR FUNDS, THREE RULES

Fund	What ₹1 lakh does to (budget, return, exposure)	Note
Nifty	(1, 12, 1)	12% return, beta 1
Nifty ETF	(1, 12, 1)	a twin of Nifty from another fund house
Bonds	(1, 7, 0)	7% return, beta 0
Balanced	(1, 9.5, 0.5)	half Nifty, half bonds

A trade is four amounts $x = (x_1, x_2, x_3, x_4)$: ₹ lakh into Nifty, ETF, bonds, balanced. Positive = buy, negative = sell.

Each fund is a column, each rule is a row:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix} \begin{array}{l} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{array}$$

The question: which trades x change nothing, so that $Ax = 0$? (That set is the nullspace $N(A)$.)

A guess

Just by looking at the menu, two do-nothing trades jump out:

1 GUESS 1

The ETF is a copy of Nifty. **Buy ₹1 lakh ETF, sell ₹1 lakh Nifty:**
 $(-1, 1, 0, 0)$.

CHECK

Budget $-1 + 1 = 0$. Return $-12 + 12 = 0$. Exposure $-1 + 1 = 0$.
 Nothing changes ✓.

2 GUESS 2

Balanced is half Nifty, half bonds. **Buy ₹1 lakh balanced, sell ₹0.5 lakh Nifty and ₹0.5 lakh bonds:** $(-0.5, 0, -0.5, 1)$.

CHECK

Budget $-0.5 - 0.5 + 1 = 0$. Return $-6 - 3.5 + 9.5 = 0$. Exposure $-0.5 + 0 + 0.5 = 0$ ✓.

But guessing leaves two questions open. **Are there other do-nothing trades we missed?** And what do we do with a menu where nothing is obvious? The recipe answers both. Watch for these two guesses: the recipe will produce them on its own.

2. Move 1: simplify the rules

Three rules about four funds are hard to read. Elimination rewrites them in a simpler form. First, let's see why that's allowed.

Each row is a question about the trade

Row	The question it asks	Its formula
1	How much does the budget change?	$x_1 + x_2 + x_3 + x_4$
2	How much does the return change?	$12x_1 + 12x_2 + 7x_3 + 9.5x_4$
3	How much does the exposure change?	$x_1 + x_2 + 0x_3 + 0.5x_4$

A do-nothing trade is one where **every answer is 0**.

Why rewriting the questions is safe

Elimination replaces a question by a combination of questions, like "return minus $12 \times$ budget". If a trade answers 0 to both "return" and "budget", it answers $0 - 12 \times 0 = 0$ to the new question. And the step can be undone (return = new question + $12 \times$ budget), so no new trade can pass that didn't pass before.

RULE · ROW OPERATIONS KEEP THE NULLSPACE

Subtracting a multiple of one row from another, swapping rows, or dividing a row by a non-zero number never changes which trades give all zeros. So $Ax = 0$ and the simplified $Ux = 0$ have exactly the same solutions. (They do change the *other* answers, so never read reachable targets off U : that's Lesson 8's job.)

Doing it: three row operations

1 WHAT

Row 2 $- 12 \times$ row 1.

ENTRIES

$12 - 12(1) = 0$, $12 - 12(1) = 0$, $7 - 12(1) = -5$, $9.5 - 12(1) = -2.5$. New row 2: $(0, 0, -5, -2.5)$.

MEANING

The new question is "how much return **beyond** what the same money would earn in Nifty at 12%?" Nifty and its ETF: 0 extra. Bonds: 5 less. Bonded: 2.5 less.

2 WHAT

Row 3 $- 1 \times$ row 1.

ENTRIES

$1 - 1 = 0$, $1 - 1 = 0$, $0 - 1 = -1$, $0.5 - 1 = -0.5$. New row 3: $(0, 0, -1, -0.5)$.

MEANING

"Exposure minus budget": how much less exposure than an all-Nifty trade of the same size.

3 WHAT

Row 3 $- 0.2 \times$ row 2 (multiplier = $(-1)/(-5) = 0.2$).

ENTRIES

$0 - 0.2(0) = 0$, $0 - 0.2(0) = 0$, $-1 - 0.2(-5) = -1 + 1 = 0$, $-0.5 - 0.2(-2.5) = -0.5 + 0.5 = 0$. New row 3: $(0, 0, 0, 0)$.

MEANING

Row 3 now asks a question whose answer is **always** 0. For these funds, exposure carries no information that budget and return don't already give.

METHOD · ECHELON FORM U (THE STAIRCASE)

Eliminate column by column, as in Lesson 2. The result U is a staircase: each row starts further right than the row above, and rows that became all zeros sit at the bottom.

$$U = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & -5 & -2.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{row 1: } x_1 + x_2 + x_3 + x_4 = 0 \\ \text{row 2: } -5x_3 - 2.5x_4 = 0 \\ \text{row 3: } 0 = 0 \end{array}$$

Same do-nothing trades as A , but now only **two** real rules, and the second one doesn't involve Nifty or the ETF at all. That's what makes the next move easy.

IN ONE LINE

Elimination rewrites the rules in a simpler form without changing which trades pass them.

3. Move 2: pivot columns and free columns

Move 1 left us with just two real rules. Written out as sentences about the trade $(x_1, x_2, x_3, x_4) =$ (Nifty, ETF, bonds, balanced):

Rule	Equation	In words
Rule 1 (row 1 of U)	$x_1 + x_2 + x_3 + x_4 = 0$	the budget doesn't change
Rule 2 (row 2 of U)	$-5x_3 - 2.5x_4 = 0$	no return beyond 12%

Two rules, four amounts. Two rules can pin down two amounts, so **two amounts are left for you to choose**. Let's see which two, by playing it as a game.

Step 1: you pick two amounts, the rules decide the rest

Say you decide: buy ₹3 lakh of the **ETF** ($x_2 = 3$) and ₹2 lakh of **balanced** ($x_4 = 2$). Now use the rules, bottom one first:

1 RULE 2

It only involves bonds and balanced: $-5x_3 - 2.5(2) = 0 \Rightarrow -5x_3 = 5 \Rightarrow x_3 = -1$. You **must** sell ₹1 lakh bonds.

RULE 1

Now only Nifty is unknown: $x_1 + 3 + (-1) + 2 = 0 \Rightarrow x_1 = -4$. You **must** sell ₹4 lakh Nifty.

2 THE TRADE

CHECK

$$(-4, 3, -1, 2).$$

Budget $-4 + 3 - 1 + 2 = 0 \checkmark$. Return $-48 + 36 - 7 + 19 = 0 \checkmark$.

Exposure $-4 + 3 + 0 + 1 = 0 \checkmark$. It changes nothing.

Pick any other ETF and balanced amounts and the same two steps always work: rule 2 hands you the bonds amount, then rule 1 hands you the Nifty amount. In general:

$$\text{bonds } x_3 = -0.5 x_4$$

$$\text{Nifty } x_1 = -x_2 - x_3 - x_4 = -x_2 - 0.5 x_4$$

Solving the bottom rule first and working upward like this is called **back substitution** (Lesson 2).

Step 2: why ETF and balanced are the ones you pick

Look at where each rule **starts**, its first non-zero entry in U . That entry is the rule's **pivot**, and it tells you which amount the rule solves for:

$$U = \begin{bmatrix} \boxed{1} & 1 & 1 & 1 \\ 0 & 0 & \boxed{-5} & -2.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rule	Starts at (pivot)	So it solves for
Rule 1	column 1	Nifty x_1
Rule 2	column 3	bonds x_3
(row 3)	nowhere: all zeros	nothing

No rule starts at column 2 or column 4, so **no rule ever solves for the ETF or balanced**. Their amounts are never decided for you. That's why they're yours to pick.

RULE · PIVOT VARIABLES AND FREE VARIABLES

A column where a rule starts is a **pivot column**; its amount is a **pivot variable**, solved for by back substitution (here Nifty x_1 , bonds x_3). A column where no rule starts is a **free column**; its amount is a **free variable**, which you choose (here ETF x_2 , balanced x_4).

Step 3: what it means for the funds

The free funds are exactly the ones that **add nothing new**, so their effect can always be undone with the pivot funds:

You buy	Why it can be undone	To undo it
₹3 lakh ETF	it is a copy of Nifty	sell ₹3 lakh Nifty
₹2 lakh balanced	it is half Nifty, half bonds	sell ₹1 lakh Nifty and ₹1 lakh bonds
Both		sell ₹4 lakh Nifty and ₹1 lakh bonds

That is exactly the trade $(-4, 3, -1, 2)$ the rules produced in step 1. The algebra and the common sense agree. Nifty and bonds can't be undone this way by the funds before them (each brings something new), so they are the ones the rules solve for.

▼ For the curious: could Nifty be free and the ETF be the pivot instead?

Yes: the twins are interchangeable, so you could just as well choose the Nifty amount and let the rules force the ETF. The recipe avoids that choice by always reading **left to right**: a column gets a pivot only if it brings something new compared with the columns *before* it. Since Nifty comes first, Nifty gets the pivot and the ETF (which repeats it) is free. Any choice gives the same set of do-nothing trades in the end.

Step 4: counting the pivots: the rank

DEFINITION · RANK r = THE NUMBER OF PIVOTS

Here two rules start somewhere, so $r = 2$. Two ways to read that number: only **two funds bring something new** (Nifty-like and bond-like), and only **two of the three rules carried information** (exposure was already decided by the other two).

And the count of free amounts follows: 4 funds minus 2 pivots = 2 free amounts to choose.

IN ONE LINE

Each rule solves for the fund where it starts (a pivot); funds where no rule starts are free for you to choose, and they're the ones that add nothing new.

4. Move 3: the special solutions

Move 2 says: choose the ETF and balanced amounts, and the rest follows. The simplest choices are to switch on **one free fund at a time**.

METHOD · SPECIAL SOLUTIONS

For each free variable in turn: set it to 1, set every other free variable to 0, and work out the pivot variables. Each free column gives one special solution.

1 CHOOSE

ETF $x_2 = 1$, balanced $x_4 = 0$.

BONDS

$$x_3 = -0.5x_4 = -0.5(0) = 0.$$

NIFTY

$$x_1 = -x_2 - 0.5x_4 = -1 - 0 = -1.$$

RESULT

$s_1 = (-1, 1, 0, 0)$. That's **guess 1**: swap Nifty for its twin.

2 CHOOSE

ETF $x_2 = 0$, balanced $x_4 = 1$.

BONDS

$$x_3 = -0.5(1) = -0.5.$$

NIFTY

$$x_1 = -0 - 0.5(1) = -0.5.$$

RESULT

$s_2 = (-0.5, 0, -0.5, 1)$. That's **guess 2**: build the balanced fund yourself.

Full check in the original rules, entry by entry:

	Budget 1, 1, 1, 1	Return 12, 12, 7, 9.5	Exposure 1, 1, 0, 0.5
$s_1 = (-1, 1, 0, 0)$	$-1 + 1 + 0 + 0 = 0 \checkmark$	$-12 + 12 + 0 + 0 = 0 \checkmark$	$-1 + 1 + 0 + 0 = 0 \checkmark$
$s_2 = (-0.5, 0, -0.5, 1)$	$-0.5 + 0 - 0.5 + 1 = 0 \checkmark$	$-6 + 0 - 3.5 + 9.5 = 0 \checkmark$	$-0.5 + 0 + 0 + 0.5 = 0 \checkmark$

The recipe found our two guesses without any guessing. On a menu where nothing is obvious, it works exactly the same way.

IN ONE LINE

One free amount = 1, the others = 0, solve for the rest: one special solution per free column.

5. Move 4: every do-nothing trade

Now the open question from topic 1: are there any others?

1 ANY CHOICE

Choose any ETF amount c and any balanced amount d . Move 2 forces $x_1 = -c - 0.5d$ and $x_3 = -0.5d$.

SPLIT IT

$$\begin{bmatrix} -c - 0.5d \\ c \\ -0.5d \\ d \end{bmatrix} = c \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -0.5 \\ 0 \\ -0.5 \\ 1 \end{bmatrix} = c s_1 + d s_2.$$

2 SO

Every do-nothing trade is $c s_1 + d s_2$: "do guess 1, c times, and guess 2, d times".

NOTHING ELSE

A do-nothing trade is pinned down by its ETF and balanced amounts. If both are 0, the forced amounts are 0 too, so the trade is zero. There's no room for a hidden third kind.

3 EXAMPLE

Someone hands you the trade $(-3, 2, -1, 2)$. Its ETF amount is 2, its balanced amount is 2, so it must be $2s_1 + 2s_2 = (-2 - 1, 2, -1, 2) = (-3, 2, -1, 2) \checkmark$.

CHECK

Budget $-3 + 2 - 1 + 2 = 0$. Return $-36 + 24 - 7 + 19 = 0$.
Exposure $-3 + 2 + 0 + 1 = 0 \checkmark$.

Two free choices give two independent directions, so $N(A)$ is a **plane** inside $\mathbb{R}^4$ (the space of four-fund trades).

Try it: choose the free amounts

ETF amount $x_2 = c$

Balanced amount $x_4 = d$

Free (you chose): ETF $x_2 = 2$, balanced $x_4 = 2$.

Forced: Nifty $x_1 = -c - 0.5d = -3$, bonds $x_3 = -0.5d = -1$.

Trade $= 2 s_1 + 2 s_2 = (-3, 2, -1, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(-3) + 1(2) + 1(-1) + 1(2) = 0 \checkmark$

Return: $12(-3) + 12(2) + 7(-1) + 9.5(2) = 0 \checkmark$

Exposure: $1(-3) + 1(2) + 0(-1) + 0.5(2) = 0 \checkmark$

Nothing changes: every rule stays at 0, whatever free amounts you pick.

IN ONE LINE

Every do-nothing trade is a mix of the special solutions, and every mix is one: $N(A) = \text{all } c s_1 + d s_2$.

6. The same four moves on smaller menus

The recipe doesn't depend on the menu. Here it is on a 2×2 and two 3×3 s, move by move.

2×2 : Nifty and its twin, budget and return only

- | | | |
|----------|---------------|--|
| 1 | MOVE 1 | $\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix}$. Row 2 $- 12 \times$ row 1: $12 - 12 = 0$, $12 - 12 = 0$. $U = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. |
| | MOVE 2 | Pivot in column 1 (Nifty); column 2 (ETF) is free. Rank 1. |
| | MOVE 3 | ETF = 1: row 1 gives $x_1 + 1 = 0$, so $x_1 = -1$. Special solution $(-1, 1)$. |
| | MOVE 4 | All do-nothing trades: $c(-1, 1)$, a line. |

3×3 : Nifty, bonds, balanced (Lesson 6's menu)

- | | | |
|----------|---------------|---|
| 1 | MOVE 1 | Same three row operations as topic 2 (just without the ETF column):
$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & 0 \end{bmatrix}$. |
| | MOVE 2 | Pivots in columns 1 and 2 (Nifty, bonds); column 3 (balanced) is free. Rank 2. |
| | MOVE 3 | Balanced = 1. Row 2: $-5x_2 - 2.5(1) = 0 \Rightarrow x_2 = -0.5$. Row 1: $x_1 - 0.5 + 1 = 0 \Rightarrow x_1 = -0.5$. Special solution $(-0.5, -0.5, 1)$. |
| | MOVE 4 | All: $c(-0.5, -0.5, 1)$, a line. ($c = -2$ gives Lesson 6's $(1, 1, -2)$.) |

3×3 : Nifty, bonds, midcap

- | | | |
|----------|------------------|--|
| 1 | MOVE 1 | Row 2 $- 12 \times$ row 1: $(0, -5, 15 - 12 = 3)$. Row 3 $-$ row 1: $(0, -1, 1.2 - 1 = 0.2)$. Row 3 $- 0.2 \times$ row 2: $(0, 0, 0.2 - 0.6 = -0.4)$. $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & 0 & -0.4 \end{bmatrix}$. |
| | MOVE 2 | Pivots in all three columns. No free column. Rank 3. |
| | MOVES 3-4 | Nothing to choose, so the only do-nothing trade is $(0, 0, 0)$. Midcap brings something new; no fund repeats another. |

How many special solutions?

Every column is either a pivot column or a free column, and each free column gives one special solution. So:

RULE · $n - r$ SPECIAL SOLUTIONS

With n funds (columns) and rank r : number of free columns = $n - r$ = number of special solutions. Since there can't be more pivots than rows, $r \leq m$. So if there are **more funds than rules** ($n > m$), at least one column is free and a do-nothing trade **always** exists.

Menu	m rules	n funds	rank r	$n - r$	Do-nothing trades
Nifty, twin (2×2)	2	2	1	1	a line
Nifty, bonds, balanced	3	3	2	1	a line
Nifty, bonds, midcap	3	3	3	0	only 0
Nifty, ETF, bonds, balanced	3	4	2	2	a plane

WHY A DESK CARES

Three things you measure and four or more products: there is **guaranteed** to be a way to switch between products that changes nothing you measure. That's where a desk looks for the cheaper route, and also where risks you *don't* measure can slip in unnoticed.

▼ Check yourself (MIT recitation): all solutions of $x - 5y + 2z = 0$.

Move 1: one row, already a staircase. Move 2: pivot in column 1 (x); y and z are free; rank 1, so $3 - 1 = 2$ special solutions. Move 3: $y = 1, z = 0$ gives $x = 5$, so $(5, 1, 0)$; $y = 0, z = 1$ gives $x = -2$, so $(-2, 0, 1)$. Move 4: all $c(5, 1, 0) + d(-2, 0, 1)$, a plane.

IN ONE LINE

Same four moves every time; the number of special solutions is funds minus pivots, $n - r$.

7. The shortcut: the reduced form R

In Move 2 we still had to solve row 2, then put that into row 1. If we push elimination two steps further, that solving is already done for us.

METHOD · REDUCED ROW ECHELON FORM R

Start from U . Divide each pivot row by its pivot (so every pivot becomes 1), then clear the entries **above** each pivot too (Gauss-Jordan, Lesson 3). The result is $R = \text{rref}(A)$: same do-nothing trades, simplest possible rules.

1 PIVOT 2 $\rightarrow$ 1

Row 2 $\div (-5)$: $0, 0, -5/(-5) = 1, -2.5/(-5) = 0.5$. Row 2 = $(0, 0, 1, 0.5)$.

CLEAR ABOVE

Row 1 has a 1 above that pivot. Row 1 $- 1 \times$ row 2: $1 - 0 = 1, 1 - 0 = 1, 1 - 1 = 0, 1 - 0.5 = 0.5$. Row 1 = $(1, 1, 0, 0.5)$.

$$R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 \\ 0 & 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{row 1: } x_1 + x_2 + 0.5x_4 = 0 \Rightarrow x_1 = -x_2 - 0.5x_4 \\ \text{row 2: } x_3 + 0.5x_4 = 0 \Rightarrow x_3 = -0.5x_4 \end{array}$$

Each row now contains **one** pivot amount and only free amounts: it is already the Move 2 answer. So the special solutions can be read off directly:

1 READING RULE

Set one free amount to 1. Each pivot amount is **minus** the number in that free column of R , in the pivot's row.

ETF = 1

ETF column of R is $(1, 0)$ in rows 1, 2. So Nifty = -1 , bonds = 0 : $s_1 = (-1, 1, 0, 0) \checkmark$.

BALANCED = 1

Balanced column of R is $(0.5, 0.5)$. So Nifty = -0.5 , bonds = -0.5 : $s_2 = (-0.5, 0, -0.5, 1) \checkmark$.

METHOD · NULLSPACE MATRIX N

Put the special solutions side by side as columns. N holds every do-nothing trade at once ($N \begin{bmatrix} c \\ d \end{bmatrix} = c s_1 + d s_2$), and $AN = 0, RN = 0$.

$$N = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \\ 0 & -0.5 \\ 0 & 1 \end{bmatrix} \quad \begin{array}{l} \leftarrow \text{Nifty} \\ \leftarrow \text{ETF} \\ \leftarrow \text{bonds} \\ \leftarrow \text{balanced} \end{array}$$

Check $RN = 0$, each row of R against each column of N :

	$s_1 = (-1, 1, 0, 0)$	$s_2 = (-0.5, 0, -0.5, 1)$
Row 1 of R : $(1, 1, 0, 0.5)$	$-1 + 1 + 0 + 0 = 0 \checkmark$	$-0.5 + 0 + 0 + 0.5 = 0 \checkmark$
Row 2 of R : $(0, 0, 1, 0.5)$	$0 + 0 + 0 + 0 = 0 \checkmark$	$0 + 0 - 0.5 + 0.5 = 0 \checkmark$
Row 3 of R : $(0, 0, 0, 0)$	$0 \checkmark$	$0 \checkmark$

▼ For the curious: Strang's compact form $R = [I \ F]$ and $N = [-F; I]$

List the pivot columns first (Nifty, bonds) and the free columns after (ETF, balanced). Then the pivot part of R is the identity $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and the free part is $F = \begin{bmatrix} 1 & 0.5 \\ 0 & 0.5 \end{bmatrix}$. The reading rule above says the pivot amounts are $-F$ and the free amounts are I , so in that order $N = \begin{bmatrix} -F \\ I \end{bmatrix}$, and $RN = I(-F) + F I = 0$.

MIT's lecture matrix $\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$ has the same shape as our menu: $R = \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, pivots in columns 1 and 3, special solutions $(-2, 1, 0, 0)$ and $(2, 0, -2, 1)$.

▼ Check yourself (MIT problem 7.1): find R , the rank and the special solutions of $\begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 2 & -2 & 11 & -3 \end{bmatrix}$.

Move 1: row 3 $- 2 \times$ row 1 $= (0, -12, -3, -21)$; row 3 $+ 3 \times$ row 2 $= (0, 0, 0, 0)$. Shortcut: row 2 $\div 4 = (0, 1, \frac{1}{4}, \frac{7}{4})$; row 1 $- 5 \times$ row 2 $= (1, 0, \frac{23}{4}, \frac{1}{4})$. So $R = \begin{bmatrix} 1 & 0 & 23/4 & 1/4 \\ 0 & 1 & 1/4 & 7/4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, rank 2, free columns 3 and 4. Reading rule: $(-\frac{23}{4}, -\frac{1}{4}, 1, 0)$ and $(-\frac{1}{4}, -\frac{7}{4}, 0, 1)$. Try it in the lab on the Examples tab.

IN ONE LINE

R is U finished off: each row says "pivot amount $= -(\text{free amounts} \times \text{numbers})$ ", so the special solutions are read straight off it.